

Likelihood

Today's agenda:

- Midterm explained
- Likelihood

Midterm

- Due October 14th by the end of the day (11:59 pm **firm deadline**)
- Take-home
- Open everything (but no AI for written components)
- Submit via canvas (file or Github link to a file)

Midterm

3 parts:

1) Writing about analyses

- a) Draft part of a short paper
- b) Use data of your choice
- c) Builds heavily on assignment 1

2) Identify which mathematical functions and distributions to use

- a) Given 3 datasets, decide which function and distribution to use for each. Explain why.

3) Power analysis to understand how sample size and slope impact power

- a) Builds off of what we did in class.
- b) I provide code to help get you started.
- c) Possible solutions range from pretty easy to more complicated.

Midterm

In part 1 I ask you to cite R and any packages used.
Here's how to get that information:

```
# Citing R
```

```
citation()
```

```
# Citing an R package
```

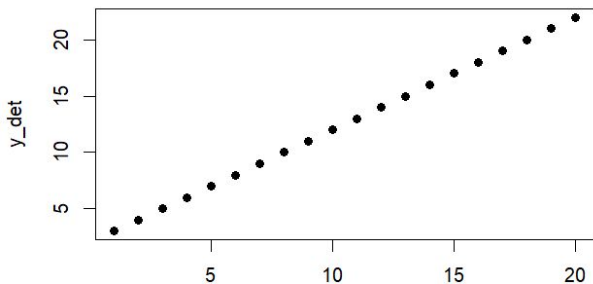
```
citation("bbmle")
```

Midterm

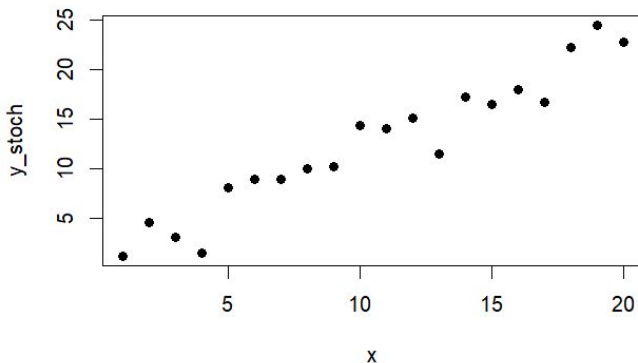
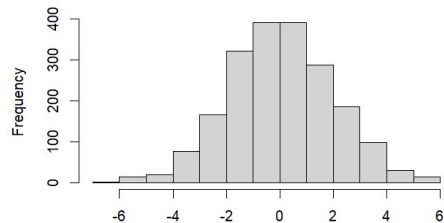
Any questions?

Last time: going from parameters to data

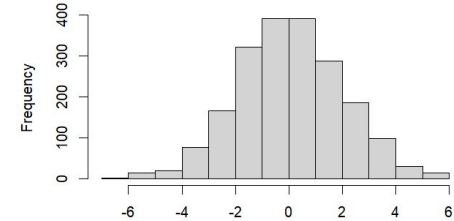
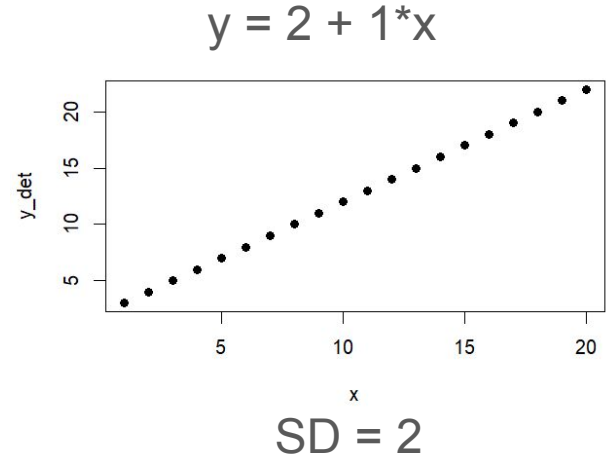
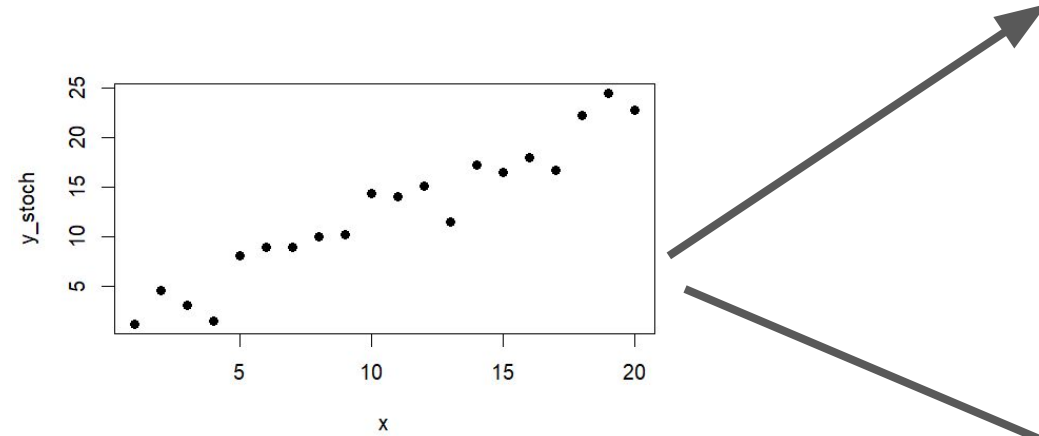
$$y = 2 + 1 \cdot x$$



SD = 2



Now: Going from data to parameters



Why do we estimate parameters from data?

In most cases, we have data but what we want to know are the parameters, e.g.:

- Slope:
 - how much does body size change with latitude?
 - How does species richness change with depth?
- Intercepts:
 - Do some species consume more than others?
 - Are some species usually more common than others?
- Variation:
 - Does variation in egg size change with body size?
 - Do leaves vary more within a given species at high latitude?

How do we estimate parameters from data?

- 1) Select models
- 2) Adjust the model parameters to maximize likelihood
- 3) Figure out which model(s) fit the data best

How do we estimate parameters from data?

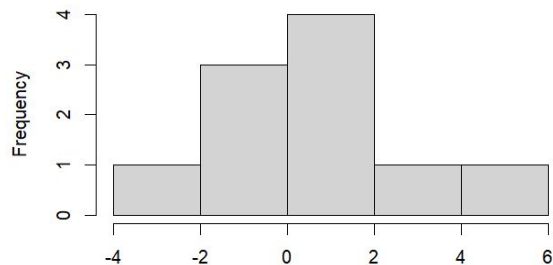
- 1) Select models **[We've done this]**
- 2) Adjust the model parameters to maximize likelihood **[This week]**
- 3) Figure out which model(s) fit the data best **[Next week]**

What is likelihood?

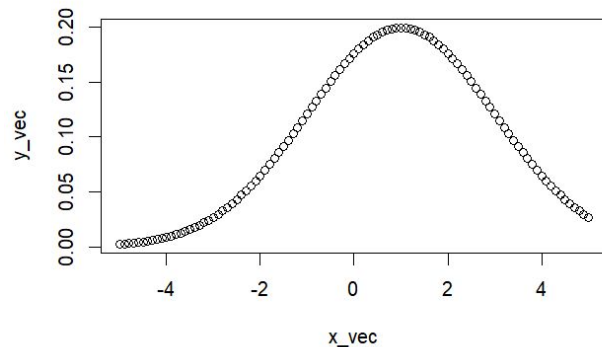
In this case:

The probability of seeing the data we collected, given a particular model.

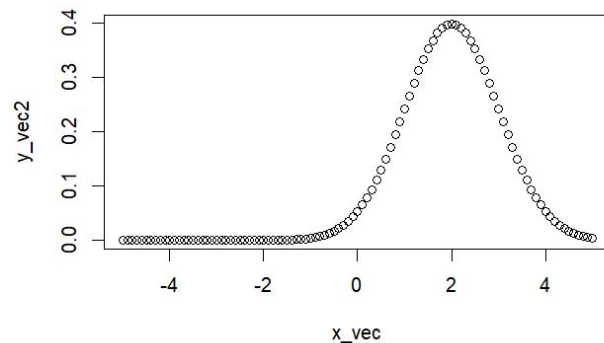
Observed



Model A



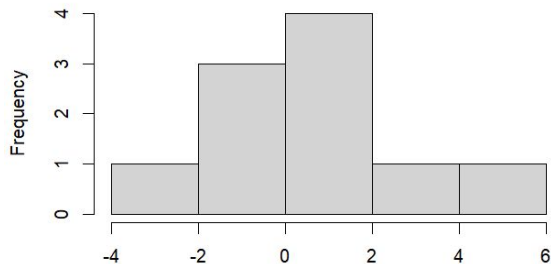
Model B



How do we maximize likelihood?

Given data and a model, we find the parameters that give us the highest likelihood

Observed



Normal distribution with:

Mean = ?

Variance = ?

Example: Tadpole data

Load the data on tadpole predation from the emdbook package

Contains information on number of tadpoles initially and the number that survived

```
library(emdbook)
```

```
data("ReedfrogPred")
```

Example: Tadpole data

To simplify things, we're going to start with a single treatment:

- Small individuals
- Stocking density of 10 tadpoles per tank

How might we narrow down the data to only this subset?

Subsetting

Use the `subset()` function

```
subset(x, subset, select, drop = FALSE, ...)
```

Subsetting

```
x <- subset(x = ReedfrogPred,  
            subset = density == 10 &  
              pred == "pred" &  
              size=="small")
```

Check to make sure x only has the expected subset

Functions

Now that we have data, we need a **function** to optimize.

Functions in R

- Lots of them built in (e.g., `lm()`, `mean()`, `plot()`)
- Others in packages (e.g., `ggplot()`, `readxl()`)
- Can easily write your own! (super helpful)

Functions

Structure of a function in R:

```
function_name <- function(x,y){  
    output <- some_operation_using_variables(x,y)  
    return(output)  
}
```

Then you use it like any other function:

```
function_results <- function_name(x = 1, y = 2)
```

Function example

```
calculate_mean <- function(x){  
  out <- sum(x)/length(x)  
  return(out)  
}
```

Should give the same results as R's mean function:

```
calculate_mean(x = c(1,2,3))  
mean(c(1,2,3))
```

Negative log likelihood function

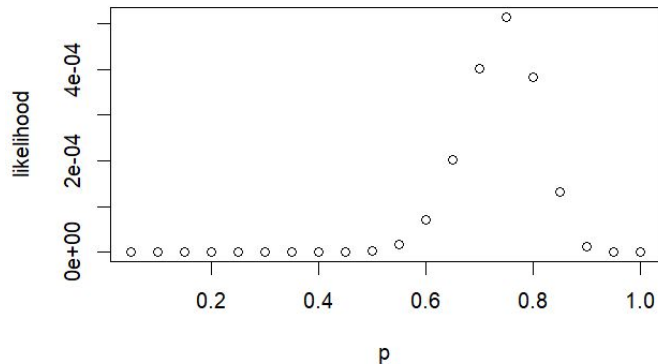
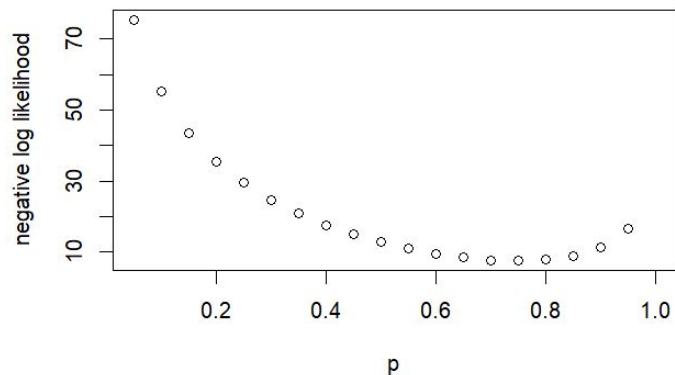
R has ways of optimizing the values in functions

However, to use them we need a function to optimize:

```
binomNLL1 <- function(p,k,N){  
  -sum(dbinom(x = k,  
              prob = p,  
              size = N,  
              log = TRUE)  
)  
}
```

Optimization

- Brute force option: try a bunch of numbers and pick the best one
- Algorithms are usually a better option. R has many options
- The shape of the NLL surface makes it easier to optimize



Optimizing using optim

`optim()` handles optimization/minimization and is used for model fitting:

```
opt1 <- optim(fn = binomNLL1,  
             par = c(p = 0.5), # starting parameter estimate  
             N = x$density, # individuals in each trial  
             k = x$surv, # survival rate  
             method = "BFGS")
```

optim results

```
opt1$par # the optimal parameter estimate for predation prob p  
opt1$value # negative log likelihood
```

To get likelihood, make it positive and exponentiate:

```
exp(-opt1$value)
```

Try this out and see what values you got

Optimization using mle2

mle2 is a function by Ben Bolker designed to be more intuitive

```
library(bbmle)
```

```
m1 <- mle2(minuslogl = binomNLL1,  
           start = c(p=0.5), #starting parameters  
           method = "BFGS",  
           data = list(N= 10, k = k) #individuals pertrial (N) and survival (k)  
           )
```

```
m1@coef # estimated parameters (p)
```

```
exp(-m1@min) # maximum likelihood
```


Remainder of Class:

- Work through the example in 6.2.1.2

Before next class:

- Read 6.3